

SAFETY DATA SHEET

ECOLAB®

INSPEXX 200

1. Chemical product and company identification

Chinese name : 雙重過酸殺菌劑
Product name : INSPEXX 200
Other names : None.
Product code : 900177-02
Recommended use and restrictions : Food additive.
Use only for the purpose on the product label.
Manufacturer/Supplier : Ecolab Limited
Address/Telephone : 6F, No. 13, Sec. 2, Peitou Rd., Peitou Dist.,
Taipei, 112, Taiwan, R.O.C.
(02) 2898-4800
Emergency telephone number : 1-651-222-5352 (in USA)

2. Hazards identification

GHS Classification : ORGANIC PEROXIDES - Type F
CORROSIVE TO METALS - Category 1
ACUTE TOXICITY: ORAL - Category 4
SKIN CORROSION/IRRITATION - Category 1B
SERIOUS EYE DAMAGE/ EYE IRRITATION - Category 1
AQUATIC TOXICITY (ACUTE) - Category 3

GHS label elements

Symbol**Signal word**

: Danger

Hazard statements

: Heating may cause a fire.
May be corrosive to metals.
Harmful if swallowed.
Causes severe skin burns and eye damage.
Harmful to aquatic life.

Precautionary statements

Prevention

: Wear protective gloves. Wear eye or face protection. Wear protective clothing.
Keep away from heat, sparks, open flames and hot surfaces. - No smoking. Keep
away from clothing, incompatible materials and combustible materials. Keep only in
original container. Avoid release to the environment. Do not eat, drink or smoke
when using this product. Wash hands thoroughly after handling.

Response

: IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable
for breathing. Immediately call a POISON CENTER or physician. IF SWALLOWED:
Immediately call a POISON CENTER or physician. Do NOT induce vomiting. IF ON
SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with
water or shower. Immediately call a POISON CENTER or physician. IF IN EYES:
Immediately call a POISON CENTER or physician.

Other hazards

: None known.

3. Composition/information on ingredients

Nature of material : Mixture
Chemical properties : Food additive.

Hazardous ingredients	Concentration Range (%)	CAS number
acetic acid	20 - 50	64-19-7
peracetic acid	5 - 20	79-21-0
octanoic acid	5 - 20	124-07-2
Hydrogen peroxide	1 - 5	7722-84-1
octaneperoxoic acid	1 - 5	33734-57-5

There are no additional ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment and hence require reporting in this section.

4. First aid measures

First aid measures

- Eye contact** : In case of contact, immediately flush eyes with cool running water. Remove contact lenses and continue flushing with plenty of water for at least 15 minutes. Get medical attention immediately.
- Skin contact** : In case of contact, immediately flush skin with plenty of water for at least 15 minutes while removing contaminated clothing and shoes. Get medical attention immediately. Wash clothing before reuse. Clean shoes thoroughly before reuse.
- Inhalation** : If inhaled, remove to fresh air. If exposed person is not breathing, give artificial respiration or oxygen applied by trained personnel. Get medical attention immediately.
- Ingestion** : Rinse mouth; then drink one or two large glasses of water. Do not induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.

Most important symptoms/effects, acute and delayed

- Eye contact** : Causes serious eye damage.
- Skin contact** : Causes severe burns.
- Inhalation** : May give off gas, vapor or dust that is very irritating or corrosive to the respiratory system.
- Ingestion** : Harmful if swallowed. May cause burns to mouth, throat and stomach.

Protection of first-aiders : No action shall be taken involving any personal risk or without suitable training. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Wash contaminated clothing thoroughly with water before removing it, or wear gloves.

Notes to physician : Treat symptomatically. Contact poison treatment specialist immediately if large quantities have been ingested or inhaled.

See toxicological information (Section 11)

5. Fire-fighting measures

Suitable fire extinguishing media : Use dry chemical, CO₂, water spray (fog) or foam.

Specific hazards arising from the chemical : This material increases the risk of fire and may aid combustion. Heating may cause a fire. May re-ignite itself after fire is extinguished. Hazardous decomposition may occur. In a fire or if heated, a pressure increase will occur and the container may burst. Runoff to sewer may create fire or explosion hazard. This material is harmful to aquatic life. Fire water contaminated with this material must be contained and prevented from being discharged to any waterway, sewer or drain.

Hazardous thermal decomposition products : Decomposition products may include the following materials:
carbon dioxide
carbon monoxide

5. Fire-fighting measures

- Specific fire-fighting methods** : Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training. Move containers from fire area if this can be done without risk. Use water spray to keep fire-exposed containers cool.
- Special protective equipment for fire-fighters** : Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

6. Accidental release measures

- Personal precautions** : Initiate company's spill response procedures immediately. Keep people out of area. Put on appropriate personal protective equipment (see section 8). Eliminate all ignition sources. Use personal protective equipment as required. Do not touch or walk through spilled material.
- Environmental precautions** : Avoid contact of spilled material and runoff with soil and surface waterways.
- Methods for cleaning up** : Follow company's spill procedures. Keep people away from spill. Put on appropriate personal protective equipment (see section 8). Absorb/neutralize liquid material. Use a tool to scoop up solid or absorbed material and put into appropriate labeled container. If emergency personnel are unavailable, contain spilled material. If emergency personnel are not present, contain and collect small spillages with non-combustible absorbent material, e.g. sand, earth or other suitable material and place in a suitable container for disposal. Absorb with an inert material and transfer the spilled material and absorbent to an appropriate waste disposal container. Use a tool to scoop up solid or absorbed material and place into appropriate labeled waste container. Use a water rinse for final clean-up.

7. Handling and storage

- Handling** : Do not ingest. Do not get in eyes or on skin or clothing. Do not breathe vapor or mist. Keep away from combustible materials. Keep away from heat, sparks and flame. To avoid fire, eliminate ignition sources. Keep only in original container. Use only with adequate ventilation. Wash thoroughly after handling.
- Storage** : Keep out of reach of children. Keep container tightly closed. Keep container in a cool, well-ventilated area. Separate from reducing agents and combustible materials.
Store between the following temperatures: 40 and -15°C

8. Exposure controls/personal protection

Control parameters

Ingredient name	Exposure limits
acetic acid	Taiwan Council of Labor Affairs (Taiwan, 1/2010). STEL: 37.5 mg/m ³ 15 minute(s). STEL: 15 ppm 15 minute(s). TWA: 25 mg/m ³ 8 hour(s). TWA: 10 ppm 8 hour(s).
Hydrogen peroxide	Taiwan Council of Labor Affairs (Taiwan, 1/2010). STEL: 2.8 mg/m ³ 15 minute(s). STEL: 2 ppm 15 minute(s). TWA: 1.4 mg/m ³ 8 hour(s). TWA: 1 ppm 8 hour(s).

Biological indicator : Not available.

8. Exposure controls/personal protection

Appropriate engineering controls : Use only with adequate ventilation. If user operations generate dust, fumes, gas, vapor or mist, use process enclosures, local exhaust ventilation or other engineering controls to keep worker exposure to airborne contaminants below any recommended or statutory limits. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

Personal protection

Hygiene measures : Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Appropriate techniques should be used to remove potentially contaminated clothing. Wash contaminated clothing before reusing.

Eye protection : Use chemical splash goggles. For continued or severe exposure wear a face shield over the goggles.

Hand protection : Use chemical-resistant, impervious gloves.

Skin protection : Use synthetic apron, other protective equipment as necessary to prevent skin contact.

Respiratory protection : A respirator is not needed under normal and intended conditions of product use.

9. Physical and chemical properties

Appearance

Physical state : Liquid.

Color : Colorless

Odor : Acetic acid. [Strong]

pH : 2.4 (100%)

Melting point : Not available.

Boiling point : Not available.

Flash point : > 100°C
Product does not support combustion.

Evaporation rate (butyl acetate = 1) : Not available.

Flammability (solid, gas) : Not available.

Explosion limits : Not available.

Vapor pressure : Not available.

Vapor density : Not available.

Relative density : 1.078 (Water = 1)

Solubility : Not available.

Partition coefficient: n-octanol/water : Not available.

Auto-ignition temperature : Not available.

10. Stability and reactivity

Stability : The product is stable.

Possibility of hazardous reactions : Hazardous reactions or instability may occur under certain conditions of storage or use.
Conditions may include the following:
temperature increase
high temperature
Reactions may include the following:
hazardous decomposition
risk of causing fire

10. Stability and reactivity

Conditions to avoid	: Avoid all possible sources of ignition (spark or flame). Do not pressurize, cut, weld, braze, solder, drill, grind or expose containers to heat or sources of ignition. Avoid increased storage temperature. Drying on clothing or other combustible materials may cause fire.
Materials to avoid	: Extremely reactive or incompatible with the following materials: organic materials and alkalis. Highly reactive or incompatible with the following materials: metals. Do not mix with bleach or other chlorinated products - will cause chlorine gas.
Hazardous decomposition products	: Under normal conditions of storage and use, hazardous decomposition products should not be produced.

11. Toxicological information

Route of exposure : Skin contact, Eye contact, Inhalation, Ingestion

Symptoms

Eye contact	: Adverse symptoms may include the following: pain watering redness
Skin contact	: Adverse symptoms may include the following: pain or irritation redness blistering may occur
Inhalation	: Adverse symptoms may include the following: coughing Respiratory tract irritation
Ingestion	: Adverse symptoms may include the following: stomach pains

Acute toxicity

Eye contact	: Causes serious eye damage.
Skin contact	: Causes severe burns.
Inhalation	: May give off gas, vapor or dust that is very irritating or corrosive to the respiratory system.
Ingestion	: Harmful if swallowed. May cause burns to mouth, throat and stomach.

Toxicity data

Product/ingredient name	Result	Species	Dose	Exposure
acetic acid	LD50 Dermal	Rabbit	1060 mg/kg	-
	LD50 Oral	Rat	3310 mg/kg	-
	LC50 Inhalation Vapor	Rat	>40 mg/l	4 hours
peracetic acid	LD50 Dermal	Rat	>12000 mg/kg	-
	LD50 Oral	Rat	263 to 1026 mg/kg	-
	LD50 Oral	Mouse	210 mg/kg	-
octanoic acid	LD50 Oral	Guinea pig	10 mg/kg	-
	LD50 Dermal	Rabbit	>5000 mg/kg	-
	LD50 Oral	Rat	>2000 mg/kg	-
	LC50 Inhalation Dusts and mists	Rat	>4.6 mg/l	4 hours
Hydrogen peroxide	LD50 Dermal	Rabbit	>2000 mg/kg	-
	LD50 Oral	Rat	486 mg/kg	-

Chronic toxicity

Carcinogenicity	: No known significant effects or critical hazards.
Mutagenicity	: No known significant effects or critical hazards.
Teratogenicity	: No known significant effects or critical hazards.

11. Toxicological information

Developmental effects : No known significant effects or critical hazards.
Fertility effects : No known significant effects or critical hazards.

12. Ecological information

Ecotoxicity : This material is harmful to aquatic life.

Aquatic and terrestrial toxicity

Product/ingredient name	Test	Result	Species	Exposure
acetic acid	-	Acute LC50 75 mg/l	Fish	96 hours
peracetic acid	-	Acute LC50 13 mg/L	Fish	96 hours
	-	Acute LC50 3.3 mg/L	Fish	96 hours
octanoic acid	-	Acute LC50 22 mg/l	Fish	96 hours
Hydrogen peroxide	-	Acute EC50 1.38 mg/l	Aquatic plants	72 hours

Persistence and degradability

Product/ingredient name	Test	Result	Dose	Inoculum
Not available.				

<u>Product/ingredient name</u>	<u>Aquatic half-life</u>	<u>Photolysis</u>	<u>Biodegradability</u>
Not available.			

Bioaccumulative potential

<u>Product/ingredient name</u>	<u>LogP_{ow}</u>	<u>BCF</u>	<u>Potential</u>
acetic acid	-0.17	-	low
octanoic acid	3.05	-	low

Mobility in soil : Not available.

Other adverse effects : No known significant effects or critical hazards.

13. Disposal considerations

Disposal methods : The generation of waste should be avoided or minimized wherever possible. Significant quantities of waste product residues should not be disposed of via the foul sewer but processed in a suitable effluent treatment plant. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible. This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Vapor from product residues may create a highly flammable or explosive atmosphere inside the container. Do not cut, weld or grind used containers unless they have been cleaned thoroughly internally. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

14. Transport information

Certain shipping modes or package sizes may have exceptions from the transport regulations. The classification provided may not reflect those exceptions and may not apply to all shipping modes or package sizes.

UN number : UN3109

Proper shipping name : ORGANIC PEROXIDE TYPE F,
LIQUID (Peroxyacetic acid, Type F,
stabilized)

Class : 5.2 (8)

Packing group : -

Marine pollutant (Yes / No) : Not a pollutant.

14. Transport information

Specific precautionary : -
transport measures and
conditions

See shipping documents for specific transportation information.

15. Regulatory information

TCSCA List of toxic chemicals

<u>Listed no.</u>	<u>Series no.</u>	<u>Product name</u>	<u>RQ</u>	<u>Class</u>
Not available.				

List of chemicals for which
manufacturing or handling
is defined as "work specially
hazardous to health" : Not available.

List of chemicals reputed to
be a "threat of imminent
danger" : This product contains substances considered to be a "Threat of imminent danger":
acetic acid, Hydrogen peroxide.

LSHL Article 21 : Not available.

Safety, health and
environmental regulations
specific for the product : Not available.

References : -Regulation of Labeling and Hazard Communication of Dangerous and Harmful Materials.
-Labor Safety and Health Act.

16. Other information

History

Date of issue : 2012 . 11 . 23

Date of previous issue : 2012 . 11 . 23

Organization that prepared
the SDS : ECOLAB

Person who prepared the
SDS : Regulatory Affairs

Notice to reader

The above information is believed to be correct with respect to the formula used to manufacture the product in the country of origin. As data, standards, and regulations change, and conditions of use and handling are beyond our control, NO WARRANTY, EXPRESS OR IMPLIED, IS MADE AS TO THE COMPLETENESS OR CONTINUING ACCURACY OF THIS INFORMATION.